

Muhammad Hassan

Lead Software Engineer | AI-Augmented Product Builder | Full-Stack SaaS Architect

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PROFESSIONAL SUMMARY

Lead Software Engineer with 5+ years of experience designing scalable SaaS platforms and AI-powered automation systems. Specialized in integrating LLMs into production workflows, including AI agent architectures, vector search pipelines, and prompt-engineered automation delivering measurable business impact (50% reconciliation time reduction, 35% API latency improvement, ~1M req/sec throughput). Core stack: Node.js, Django, React, PostgreSQL, AWS. AI-first developer using GenAI tools (Cursor, GitHub Copilot, Claude) daily. Led engineering teams of 3–5, architected multi-tenant platforms, and driven strategic technical decisions across fintech, healthcare, and AI-driven SaaS domains. Strong interest in applying AI/LLM capabilities to EdTech, personalized learning, and knowledge automation.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python

Frontend: React, Redux, Context API, Next.js, HTML5, CSS3, Responsive Design

Backend: Node.js, Express, NestJS, Django, REST APIs, GraphQL

AI / LLM: LLM API Integration (GPT, Claude, Mistral), Vector Search, Prompt Engineering, AI Agent Design, RAG Pipelines, GenAI Dev Tools (Cursor, GitHub Copilot)

Databases: PostgreSQL, MySQL, MongoDB, PostGIS, Redis

Cloud / DevOps: AWS (EC2, RDS, S3, ECS, CloudWatch), Docker, Nginx, CI/CD Pipelines

Architecture: Distributed Systems, Multi-Tenant SaaS, Microservices, Data Pipelines, Message Queues, Event-Driven Architecture, Async Processing

PROFESSIONAL EXPERIENCE

Lead Software Engineer — Circler Inc. (AI-Powered Fintech SaaS, US-based, Remote)

Mar 2025 – Present

- Architected AI-powered bookkeeping automation platform integrating banking, accounting, and marketplace data using LLM-driven workflows
- Designed AI reconciliation agent using prompt engineering and LLM APIs (GPT, Claude), reducing reconciliation time by 50% across 20+ companies
- Built vector search classification pipeline processing 1GB datasets for intelligent transaction categorization
- Processed 500k-row CSV datasets with automated AI-powered matching and reconciliation within 5 minutes
- Reduced API latency ~35% via indexing, query optimization, and async AI processing pipelines
- Led engineering team (3–5 engineers) handling architecture decisions, code reviews, and production deployments
- Maintained 99.9% uptime processing 1,000+ daily financial transactions across 20+ companies

Tech: Django, React, PostgreSQL, AWS, Docker, LLM APIs, Vector Search, Prompt Engineering, Celery, Async Processing, Data Pipelines

L5 Software Engineer (Full-Time) — Red Buffer, Islamabad

Dec 2020 – Feb 2025

- Engineered high-throughput backend handling ~1M requests/sec for distributed health tracking SaaS platform
- Served as primary technical contributor driving architecture, database design, and deployment strategy
- Implemented load-balanced microservice architecture for real-time field data ingestion across Pakistan
- Designed materialized-view reporting pipelines reducing analytics query time by ~80%
- Optimized geospatial data delivery by reducing multi-GB polygon datasets to 7–9MB for real-time map rendering
- Built multi-tenant API access system with token authentication, scoped permissions, rate limiting, and usage monitoring

- Built asynchronous processing workflows using message queues for fault-tolerant data synchronization
- Improved database performance through indexing strategies, query refactoring, and connection pooling

Tech: Node.js, React, PostgreSQL, PostGIS, MongoDB, AWS, Docker, Nginx, Microservices, Message Queues, Redis

Software Engineer, Backend — YouScale LLC / iClosed (US-based SaaS)

Feb 2023 – Nov 2023 (Part-Time)

- Developed scheduling engine with automated email/SMS reminder pipelines for SaaS meeting platform
- Built cron-based automation pipelines for event-driven notifications and workflow triggers
- Managed production deployments and backend services using Node.js

Tech: Node.js, Express, REST APIs, Cron Automation

Associate Software Engineer — i2c Inc.

2020

- Monitored high-availability production systems and resolved incidents improving operational reliability
- Built automation scripts to streamline monitoring workflows, reducing manual intervention and response times

Junior MERN Developer — Click Matrix (Pvt) Ltd

2020

- Developed responsive React interfaces and RESTful APIs with Node.js/Express and MongoDB
- Participated in code reviews and feature planning, ensuring maintainable deliverables

AI & LLM INTEGRATION EXPERIENCE

- Designed and deployed AI agent workflows integrating LLM APIs (GPT, Claude) into production SaaS for automated financial reconciliation and data classification
- Built vector search and RAG classification pipelines for intelligent data categorization at scale (1GB+ datasets)
- Engineered prompt-based automation workflows that reduced manual reconciliation effort by 50%
- Implemented AI-powered data matching across 500k-row datasets with automated quality scoring and validation
- AI-first development practitioner: daily use of Cursor, GitHub Copilot, and Claude for accelerated full-stack development

EDUCATION

BS Computer & Information Sciences

Pakistan Institute of Engineering and Applied Sciences (PIEAS) | 2016 – 2020 | CGPA: 3.47 / 4.0